Session Objectives

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1. List and differentiate bacterial infections of the respiratory system associated with community acquired pneumonia.
2. Describe the unique microbiologic features and any associated virulence factors for each pathogen
3. Explain any relevant symptoms and diagnostic tests for each pathogen.
4. Describe any treatment or prevention strategy for each pathogen.

Pneumonia

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- Inflammation of the lung affecting the alveoli
  - Alveoli can become filled with fluid or pus
- Epidemiology:
  - Over 4.5 million cases per year in USA
  - Most common infectious cause of death
- Pneumonia is frequently categorized based on site of acquisition
  - Community acquired (both typical and atypical)
  - Nosocomial pneumonia
    - hospital-acquired pneumonia & ventilator-associated pneumonia
- Diagnosed by chest x-ray: “shadow” or opaque area is present over a portion of the lung
- Age and predisposing factors are important for diagnosis

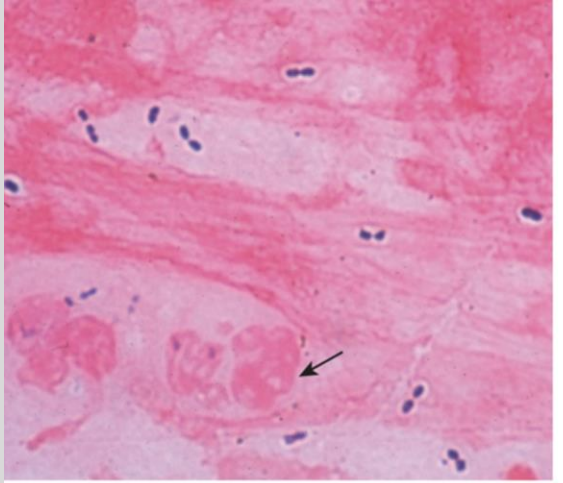
Bacterial Species

- Community-acquired pneumonia
  - *Streptococcus pneumoniae*
  - *Streptococcus pyogenes*
  - *Staphylococcus aureus*
  - *Haemophilus influenzae*
  - *Klebsiella pneumoniae*
  - *Bordetella pertussis*
- Atypical pneumonia
  - *Mycoplasma pneumoniae*
  - *Legionella pneumophila*
  - *Chlamydophila (Chlamydia) pneumoniae*

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Streptococcus pneumoniae

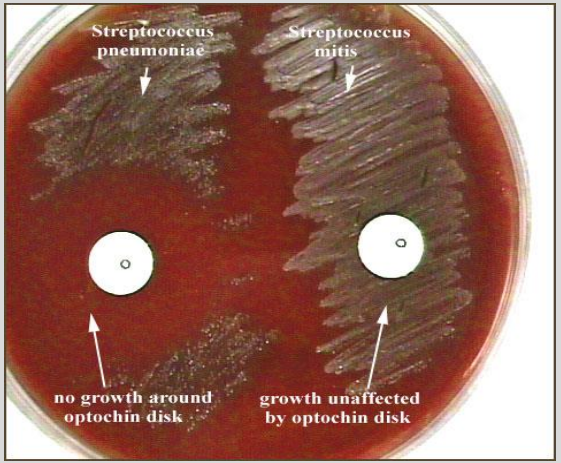
- Characteristics:
  - **Gram +ve**
  - Lancet-shaped diplococci/chains
  - Catalase –ve
  - α-hemolytic
  - Optochin sensitive and bile soluble
- Urine antigen test
- Virulence:
  - IgA protease
  - Encapsulated
    - Basis for vaccines



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e
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Most commonly causes **MOPS**:

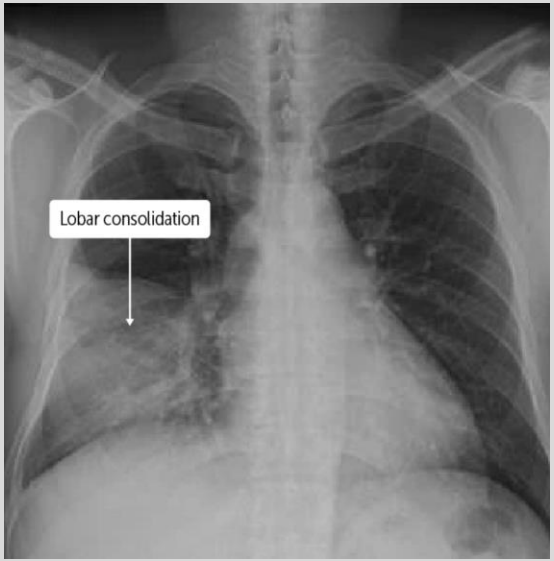
Meningitis
Otitis media
Pneumonia
Sinusitis



Streptococcus pneumoniae

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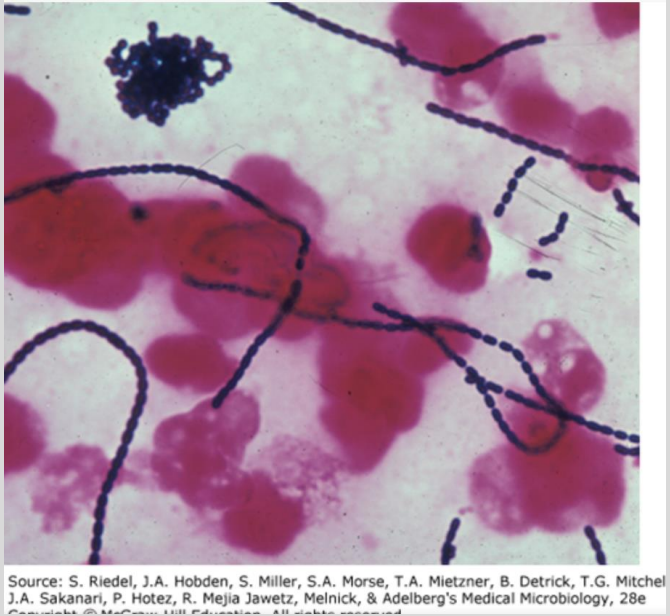
- Pneumococcal pneumonia
  - S/sx: sudden onset with fever, chest pain & thick, rusty sputum
  - Common in young-to-middle-age adults- rare but lethal in infants and elderly
  - Infection: Aerosolized bacteria are inhaled into alveoli or already colonized bacteria are aspirated into lungs
  - Classic lobar form
  - Asplenic patients are predisposed



Streptococcus pyogenes

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- Aka Group A Strep (GAS)
- Characteristics:
  - Gram +ve cocci in chains
  - Bacitracin sensitive
  - β-hemolytic
  - Catalase -ve
- Virulence:
  - Streptokinase, streptolysin S & O, DNase; Capsule contains hyaluronic acid & “M” protein (found in cell membrane & wall) and these lead to resistance to phagocytosis



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchel J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

Streptococcus pyogenes

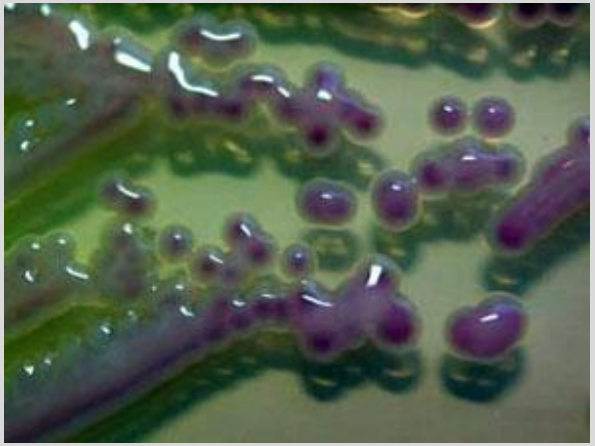
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- Transmission: respiratory droplets
- Causes
  - strep pharyngitis (which can lead to rhematic fever and glomerulonephritis), and rarely pneumonia
  - Also causes cellulitis, impetigo, erysipelas, scarlet fever
- Dx: clinical dx; rapid strep antigen test and/or culture swab on blood agar; look for hemolysis and bacitracin sensitivity; also can detect antibodies for prior infection (ASO titers)

Klebsiella pneumoniae

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- Characteristics:
  - **Gram –ve** Bacillus
  - Enteric, Non-motile, Lactose-fermenter
- Can be aspirated from GI tract and then reach respiratory tract
- Virulence:
  - Endotoxin (or LPS)
  - Polysaccharide capsule
- Causes lobar pneumonia, commonly in patients with alcohol abuse
- S/sx: fever, dyspnea “currant jelly” sputum



ABCDE’s of Klebsiella:
Aspiration pneumonia
aBscess in lungs and liver
Currant jelly sputum
Diabetes mellitus
EtOH overuse

Bordetella pertussis

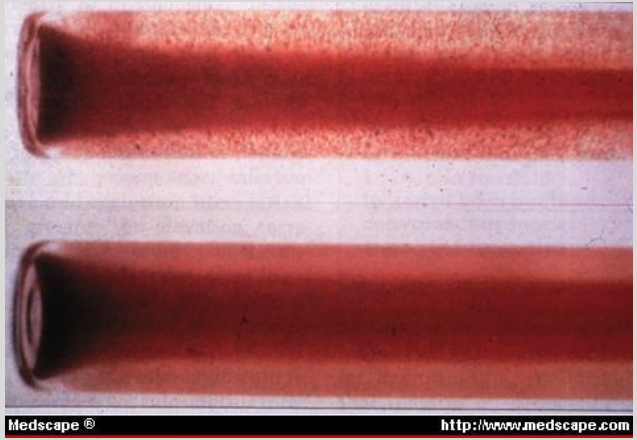
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- Characteristics:
  - Gram –ve aerobic coccobacillus
  - Grows only on Bordet-Gengou selective media
- Virulence:
  - Pertussis Toxin: Leads to edema in respiratory mucosa; Produces lymphocytosis
  - Adenylate Cyclase Toxin: Directly impairs phagocytes
  - Tracheal Cytotoxin: Kills ciliated epithelial cells
- Three clinical stages of whooping cough:
  - Catarrhal, Paroxysmal- intense cough followed by inspiratory “whoop”, Convalescent
- Prevention: Tdap and DTaP vaccines

Mycoplasma pneumoniae

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- s/sx: nonproductive cough, fever, headache, malaise, etc.
- Dx:
  - Clinical; chest X-ray shows infiltrate
  - Serology (cold agglutinins)
    - IgM antibodies cross-react with RBCs at cold temperatures → agglutination
- Tx: macrolides, doxycycline, or FQ's (non-beta lactam abx because no cell wall!)



Legionella pneumophila

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- Characteristics:
  - Gram –ve rod (stains poorly), silver stain (biopsy)
  - Grows on BCYE medium with iron and cysteine
- Transmission:
  - Aerosol from environmental water source
  - No person-to person transmission
- Legionnaires' disease:
  - Mild flu-like to severe pneumonia (with possible GI and CNS symptoms)
  - Related to Pontiac fever
- Diagnosis:
  - Detected by presence of antigen in urine
  - Labs show hyponatremia
- Tx: macrolide or quinolone

Practice Question #1

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Q: A 65-year-old man with chronic obstructive pulmonary disease presents to clinic with sudden onset fever, productive cough with rust-colored sputum, and pleuritic chest pain. Chest X-ray shows lobar consolidation. Which virulence factor of the most likely pathogen allows it to evade phagocytosis?

A. IgA protease
B. Polysaccharide capsule
C. M protein
D. Exotoxin A
E. Lipooligosaccharide

Summary Slide

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- Community-acquired pneumonia
  - *Streptococcus pneumoniae:* Gram + lancet diplococci; α-hemolytic; Catalase -; optochin sensitive; bile soluble; encapsulated
  - *Streptococcus pyogenes:* Gram + cocci chains; β-hemolytic; catalase -; bacitracin sensitive; ASO titers; hyaluronic acid and M protein virulence; strep pharyngitis
  - *Staphylococcus aureus:* Gram + cocci in clusters; β-hemolytic; catalase +; coagulase +; Prtn A
  - *Haemophilus influenzae:* Gram – rod; chocolate agar; epiglottitis
  - *Klebsiella pneumoniae:* Gram – rod; lactose fermenter on MacConkey agar; enteric; encapsulated; currant jelly sputum; alcohol use, diabetes, aspiration
  - *Bordetella pertussis:* Gram – coccobacillus; Bordet-Gengou media; pertussis toxin; 3 clinical stages
- Atypical pneumonia
  - *Mycoplasma pneumoniae:* sterol membrane; Eaton agar; cold agglutinins
  - *Legionella pneumophila:* Gram – rod (stains poorly); charcoal yeast extract medium; hyponatremia on labs; heating/cooling systems; Legionnaire's disease
  - *Chlamydophila pneumoniae:* obligate intracellular; Elementary body vs reticulate body